

Multiscale Graph Transformer for Brain Disorder Diagnosis

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Abstract—Multiscale brain networks are crucial for diagnosing brain disorders by revealing the hierarchical organization of brain function and connectivity. However, previous methods that explored multi-atlas approaches to model these networks often failed to represent this organization across multiple spatial and temporal scales, leading to limited representations and potentially inaccurate diagnoses. To address this issue, we propose the Multiscale Brain Graph Transformer (BrainMGT), which captures the hierarchical organization of brain connectivity at various spatial and temporal scales to improve the diagnosis of brain disorders. BrainMGT constructs multiscale brain networks that model spatial hierarchies, i.e., microscale, mesoscale, and macroscale, while preserving their modular structure. It also incorporates multiple temporal scales, i.e., fast, intermediate, and slow, for estimating connectivity, rather than relying on a single temporal scale from blood-oxygen-level-dependent (BOLD) signals. Using self-attention and cross-attention mechanisms, BrainMGT extracts and integrates features both within and between these scales, generating fine-coarse feature representations that improve diagnostic precision. We validated BrainMGT on three real-world functional magnetic resonance imaging (fMRI) datasets, and the results show that BrainMGT outperformed existing methods in diagnosing neurological disorders.

Index Terms—Multiscale brain networks, graph transformer, fMRI analysis, brain disorder diagnosis, brain connectivity.

I. INTRODUCTION

GRAPH theory plays a crucial role in understanding brain networks and diagnosing neurological disorders [1]. By modeling brain connectivity as a network, brain regions are represented as nodes, and their connections (either functional or structural) are represented as edges. Functional Magnetic Resonance Imaging (fMRI) is key to constructing these networks by tracking brain activity over time [2], [3]. Functional brain networks reveal interactions between brain

regions, helping identify connectivity disruptions in disorders such as Alzheimer’s disease (AD), Parkinson’s disease (PD), and autism spectrum disorder (ASD). Analyzing these networks can detect subtle connectivity changes that traditional methods may overlook, enabling earlier and more accurate diagnoses [4]. This approach enhances diagnosis, treatment, and patient outcomes through personalized therapeutic strategies.

Various methodologies have been developed for brain network analysis to aid in diagnosing neurological disorders [5], [6]. These methods are divided into single-scale and multiscale approaches [7]. Single-scale networks, typically derived from fMRI data using atlases like the Automated Anatomical Labeling (AAL) atlas and Pearson correlation, are analyzed using structural graph features, often enhanced by machine learning techniques [2], [8]. Recent advances in end-to-end graph learning methods have improved diagnostic precision [9]. However, single-scale approaches are limited in capturing brain function across varying spatial and temporal resolutions. Multiscale networks, which integrate data from multiple atlases, offer a more comprehensive analysis [10] and are processed using advanced graph learning techniques that combine information across scales [11]. Graph Transformers, leveraging self-attention mechanisms, have emerged as an effective tool for capturing complex dependencies in brain networks. These models show potential for improving diagnostic accuracy over traditional methods [12], [13], [14], [15], [16].

Multiscale brain network analysis reveals the hierarchical organization of brain structure and function, aiding the evaluation of neurological conditions [17]. However, many approaches fail to capture this hierarchy across multiple spatial and temporal scales. These methods typically rely on separate brain atlases, such as AAL116, CC200¹ (Craddock 200), and BN273² (Brainnetome), which divide the brain into regions to assess connectivity [18]. While these atlases model connectivity at specific spatial resolutions, they do not integrate data across varying spatial and temporal layers. For instance, interactions identified with BN273 may not align with broader patterns from AAL116. Additionally, these methods often focus on a single temporal scale, overlooking the brain’s dynamic nature and time-dependent interactions [19]. Given the brain’s involvement in both fast and slow processes,

Received 24 September 2024; revised 27 November 2024 and 5 January 2025; accepted 8 February 2025. Date of publication 17 February 2025; date of current version 14 August 2025. (*Corresponding author: Dongyu Zhang.*)

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Digital Object Identifier 10.1109/TCE.2025.3542794

¹https://ccraddock.github.io/cluster_roi/

²<https://atlas.brainnetome.org/bnatlas.html>

an integrated multiscale analysis is essential to accurately represent its connectivity [20], [21]. Thus, approaches that combine spatial and temporal multiscale data are needed.

The proposed Multiscale Brain Graph Transformer (BrainMGT) model addresses issues in brain connectivity analysis by incorporating hierarchical structures across multiple spatial and temporal dimensions. It constructs multiscale brain networks at microscale, mesoscale, and macroscale levels while preserving the brain's modular architecture. Connectivity is evaluated both within and between functional modules at different temporal bands (fast, intermediate, and slow), rather than relying on a single temporal scale. BrainMGT employs scale-specific encoders with self-attention mechanisms to capture intra-scale dependencies, while cross-scale encoders with cross-attention mechanisms model interactions between scales. Hierarchical adaptive fusion merges features within and across scales, generating a comprehensive representation from fine to coarse levels. Classification on three fMRI datasets (Alzheimer's Disease Neuroimaging Initiative (ADNI), Parkinson's Progression Markers Initiative (PPMI), and Autism Brain Imaging Data Exchange (ABIDE)) demonstrates that BrainMGT outperforms current state-of-the-art methods.

Below are the highlights of the work reported in this paper:

1. We propose the BrainMGT model to address the limitations of multiscale approaches in capturing the hierarchical organization of brain connectivity at multiple spatial and temporal scales for improved diagnosis of brain disorders.
2. The proposed model is designed to capture hierarchical connectivity at various spatial scales (microscale, mesoscale, macroscale) and dynamic interactions across different temporal scales (fast, intermediate, slow). BrainMGT uses self-attention and cross-attention to extract and integrate intra-scale and inter-scale features, producing a fine-coarse feature representation that improves brain disorder diagnosis accuracy.
3. Experimental results on the ADNI, PPMI, and ABIDE datasets show that the proposed BrainMGT model outperforms the baselines for differentiating between brain disorders with potential diagnostic applications.

The organization of this paper is outlined as follows: In Section II, we provide an overview of relevant literature. Section III describes the proposed methodology. Section IV reports the experimental results, discusses their significance, and considers the broader impacts. Lastly, Section V provides the conclusion of this study.

II. RELATED WORK

A. Brain Network Analysis

The study of brain networks is essential for diagnosing neurological conditions and understanding brain functionality [22]. These networks represent functional or structural interactions among brain regions and provide a graph-theoretical framework for interpreting interregional communication [23], [24]. Brain networks can be classified as single-scale or multiscale [25]. Single-scale networks

assume connectivity is captured at one spatial level [7] and are typically derived from fMRI data using predefined brain atlases, such as the Automated Anatomical Labeling (AAL) atlas [26]. Functional connections between regions are commonly assessed using Pearson correlation, resulting in a connectivity matrix [18], [27]. Traditional graph-based features like node degree and modularity are used in machine learning models such as Support Vector Machines (SVM) and Random Forests. However, single-scale approaches often fail to capture the complex dynamics of brain connectivity across spatial and temporal dimensions, limiting their diagnostic accuracy [5], [28], [29].

Multiscale brain networks account for connectivity across multiple spatial levels, from individual neurons (microscale) to larger networks (macroscale) [17]. These networks are modeled using fMRI data and a multi-atlas strategy, which captures varying spatial resolutions to generate connectivity matrices reflecting the brain's hierarchical structure [10]. Analyzing multiscale networks requires advanced methods beyond single-scale approaches. Recent work has integrated cross-scale information using graph learning algorithms. For example, Yao et al. [11] introduced the Mutual Multiscale Triplet Graph Convolutional Network (MMTGCN), combining functional and structural MRI data to improve disorder classification. Liu et al. [15] developed the Multiscale-Atlases-based Hierarchical Graph Convolutional Network (MAHGCN), enhancing diagnostic accuracy through hierarchical pooling. Lee et al. [16] proposed integrating spectral Graph Convolutional Networks (GCNs) with ensemble learning for better diagnosis. However, these methods often fail to fully capture complex cross-scale interactions, limiting their modeling of the brain's hierarchical structure.

B. Graph Transformers

Graph transformers extend transformer architectures beyond natural language processing (NLP) and computer vision to handle graph-structured datasets [30], [31]. While conventional graph neural networks (GNNs) capture local graph topologies, they struggle with long-range interactions and global context [14], [32]. In contrast, graph transformers use attention mechanisms to learn relationships among all nodes, regardless of distance or direct connections. This approach enables each node to consider any other node, capturing both local and global dependencies. As a result, graph transformers offer a more comprehensive representation of graph structures, excelling in tasks requiring integrated data understanding. Their effectiveness spans various applications, including molecular modeling, social network analysis, and recommendation systems [12], [33]. In brain disease diagnosis, they have shown significant potential in modeling complex neural interactions [34].

Despite recent advances, several issues persist in current brain network analysis methods. Most graph-based approaches rely on multi-atlas strategies, which capture brain connectivity at various scales. However, these methods often fail to account for the brain's hierarchical structure across spatial and temporal dimensions. The complex dependencies between scales

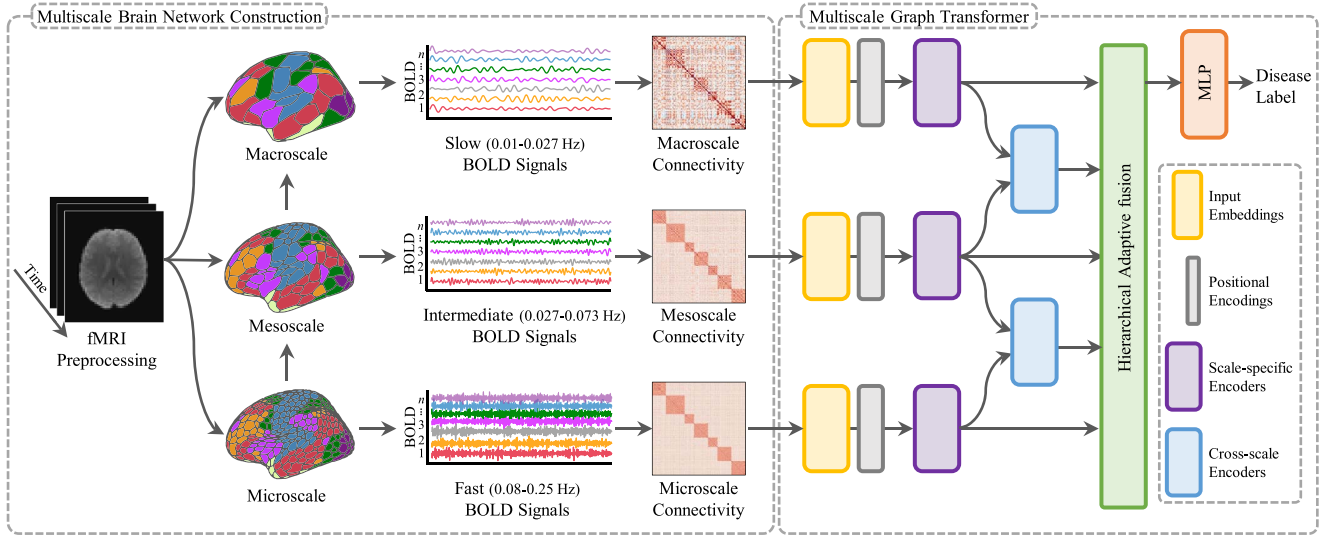


Fig. 1. The BrainMGT framework consists of two main components: (1) Multiscale Brain Network Construction, which preprocesses fMRI data, constructs networks at micro, meso, and macro scales, and analyzes connectivity across frequency bands to capture temporal dynamics; and (2) Multiscale Graph Transformer Model, which embeds, encodes, and integrates these multiscale features for accurate brain disorder classification.

have not been investigated, leading to incomplete representations of brain connectivity. This may result in missing subtle changes leading to suboptimal or inaccurate diagnoses.

III. DESIGN OF BRAINMGT

A. Overview of BrainMGT

Accurate brain network analysis is critical for diagnosing neurological disorders. Traditional methods often fail to capture hierarchical connectivity across spatial and temporal dimensions, leading to incomplete models and potential diagnostic errors. We introduce the BrainMGT framework, which improves diagnostic precision by leveraging multiscale brain networks. It consists of two main components: Multiscale Brain Network Construction and the Multiscale Graph Transformer Model. The first component constructs brain networks at microscale, mesoscale, and macroscale levels from preprocessed fMRI data, evaluating connectivity across multiple frequency bands. The second component uses graph transformers to embed, encode, and integrate these multiscale features for precise disorder classification. The architecture and workflow of BrainMGT are shown in Fig. 1.

B. Multiscale Brain Network Construction

1) *fMRI Preprocessing*: Preprocessing fMRI data is crucial for artifact reduction and dataset standardization, enabling multiscale brain network construction [18]. We apply rigid-body motion correction using FSL's MCFLIRT³ and AFNI's 3dvolreg,⁴ which align all volumes to a reference using the transformation matrix $T = \begin{bmatrix} R & t \\ 0 & 1 \end{bmatrix}$, where R is the rotation matrix and t is the translation vector. For slice timing correction, we adjust voxel time series $X(t)$ for slice-specific

delays using $X'(t) = X(t + \Delta t_s)$. Spatial normalization aligns images to a standard brain template (e.g., MNI space) through affine and nonlinear transformations, represented by the matrix A and deformation field $D(x)$. We apply Gaussian smoothing $G(x - x_i, \sigma)$ to enhance signal-to-noise ratio and reduce high-frequency noise. Finally, noise removal is performed using CompCor and ICA, regressing out noise components as $Y' = Y - N\beta$.

2) *Hierarchical Spatial Scales*: Brain architecture has been described using fractals [35], indicating the importance of defining the brain networks at multiple spatial resolutions for understanding the brain's hierarchical structure. Connectivity analysis at the microscale, mesoscale, and macroscale provides a comprehensive view of brain function [36]. At the microscale, individual neurons and synapses reveal fundamental neural processing [37]. The mesoscale focuses on neuronal assemblies and local circuits, which define connectivity within specific brain regions [38]. Macroscale studies examine large-scale networks and inter-regional interactions that support cognitive functions [8]. Connectivity abnormalities across these scales are linked to neurological disorders such as Alzheimer's disease and Autism Spectrum Disorder [39], [40]. This section outlines the methodology for constructing brain networks across these scales.

a) *Microscale*: At the microscale, we aim to capture fine-grained brain network details that reveal localized functions and connectivity patterns. We use the 7-module Schaefer atlas to define 1000 nodes from fMRI data [41]. Each node corresponds to a distinct brain region and is assigned to one of the 7 modules in the atlas. This assignment is based on the parcellation of the brain into 1000 regions, with each region linked to a specific module. Mathematically, let N_{micro} denote the set of 1000 nodes and M_7 the 7 modules. Each node $n_i \in N_{micro}$ is assigned to a module $m_j \in M_7$ such that:

$$\forall n_i \in N_{micro}, \exists m_j \in M_7 \text{ where } n_i \rightarrow m_j. \quad (1)$$

³FSL's MCFLIRT: <https://fsl.fmrib.ox.ac.uk/fsl/fslwiki/MCFLIRT>.

⁴AFNI's 3dvolreg: https://afni.nimh.nih.gov/pub/dist/doc/program_help/3dvolreg.html.

b) Mesoscale: At the mesoscale, we aim to identify neural communities of intermediate size, essential for understanding the brain's modular structure. Using the Louvain algorithm for community detection [42] on a connectivity matrix of 1,000 microscale nodes, we identify approximately 500 communities. These communities are mapped to the Schaefer atlas, with each assigned to one of seven modules based on their constituent nodes. Let C_{meso} represent the set of 500 communities. The Louvain method optimizes the modularity function Q , defined as:

$$Q = \frac{1}{2m} \sum_{i,j} \left[A_{ij} - \frac{k_i k_j}{2m} \right] \delta(c_i, c_j), \quad (2)$$

where A_{ij} is the edge weight between nodes i and j , k_i and k_j are the degrees of the nodes, m is the total number of edges, and $\delta(c_i, c_j)$ is the Kronecker delta function, equal to 1 if i and j are in the same community.

c) Macroscale: At the macroscale, we aim to capture large-scale brain network organization, providing a broad overview of connectivity. The 500 mesoscale communities are aggregated into about 100 larger communities based on connectivity patterns. These communities are then mapped to the Schaefer atlas, with each assigned to one of the 7 modules. Let C_{macro} represent the set of 100 larger communities. The aggregation process involves clustering mesoscale communities C_{meso} such that:

$$C_{\text{macro}} = \text{Aggregate}(C_{\text{meso}}). \quad (3)$$

Each larger community $c_i \in C_{\text{macro}}$ is assigned to a module $m_j \in M_7$, consistent with the microscale and mesoscale mappings. The mapping process ensures that nodes and communities are assigned to one of the 7 modules based on their spatial location and connectivity. This hierarchical assignment maintains consistent representation of brain networks at all scales within the Schaefer atlas. By defining brain networks at multiple scales, we capture the brain's complex hierarchical organization, providing a foundation for multiscale network analyses.

3) Multi-Band Temporal Scales: Analyzing brain activity at different temporal resolutions is essential for capturing dynamic changes across frequency bands [43]. Each frequency range is linked to distinct neural processes, so examining multiple temporal scales provides a more complete understanding of brain function [44]. The fast (0.08-0.25 Hz), intermediate (0.027-0.073 Hz), and slow (0.01-0.027 Hz) bands are selected based on their association with specific neural activities and clinical relevance. Understanding activity within these bands is critical for studying neural dynamics and brain disorders. This section outlines the methodology for analyzing brain activity across these frequency bands.

a) Fast frequency BOLD signals: The BOLD signals are used from fMRI data to estimate brain activation. To isolate high-frequency activity, we focus on the 0.08-0.25 Hz range, where BOLD fluctuations convey significant neural information [45]. Fluctuations in this range reflect local neuronal processing, offering insights into conditions like epilepsy, where local circuitry is affected. We filter BOLD

signals from 1000 microscale nodes using the Discrete Wavelet Transform (DWT), which provides effective time-frequency localization for capturing dynamic brain activity [46]. The DWT of a signal $x(t)$ is given by:

$$W_\psi(a, b) = \frac{1}{\sqrt{|a|}} \int_{-\infty}^{\infty} x(t) \psi\left(\frac{t-b}{a}\right) dt, \quad (4)$$

where $W_\psi(a, b)$ are the wavelet coefficients, a is the scaling factor, b is the translation factor, and ψ is the wavelet function. We use the Daubechies wavelet due to its compact support and efficiency in capturing transient features in BOLD signals.

b) Intermediate frequency BOLD signals: We analyze Intermediate Frequency Band BOLD signals in the 0.027-0.073 Hz range to capture mid-frequency brain activity. This range is linked to cognitive functions and regional brain interactions [43]. Intermediate frequencies are essential for studying cognitive processes, such as attention and memory. Abnormalities in this band are associated with cognitive disorders, making it relevant for clinical studies. To isolate this frequency range, we filter BOLD signals from 500 mesoscale communities using the Discrete Wavelet Transform (DWT), which efficiently decomposes the signals.

c) Slow frequency BOLD signals: We analyze Slow Frequency Band BOLD signals in the 0.01-0.027 Hz range to study low-frequency brain dynamics. This range is associated with long-distance neural communication and large-scale brain network coordination [47]. Slow-frequency oscillations play a key role in global brain connectivity and are linked to neurodegenerative and psychiatric disorders. Examining this band provides insights into disruptions in large-scale networks associated with these conditions. Empirical data emphasize the importance of slow oscillations in resting-state functional connectivity. We apply the Discrete Wavelet Transform (DWT) with optimized parameters to capture the relevant slow-frequency components.

4) Hierarchical Multiscale Connectivity: Constructing brain networks across multiple spatial and temporal scales is crucial for analyzing connectivity patterns and understanding brain function. This section describes the methodology for building hierarchical multiscale connectivity networks, with an emphasis on intra-module and inter-module connectivity.

a) Intra-module connectivity: Intra-module connectivity measures interaction within each module across spatial (micro, meso, macro) and temporal (fast, intermediate, slow) scales. For each scale, the Pearson correlation between BOLD signals of elements within the same module is computed. At the microscale, fast-frequency BOLD signals from 1,000 nodes are analyzed. The correlation coefficient r_{ij} between nodes i and j is defined as:

$$r_{ij} = \frac{\sum_t (x_i(t) - \mu_i)(x_j(t) - \mu_j)}{\sqrt{\sum_t (x_i(t) - \mu_i)^2 \cdot \sum_t (x_j(t) - \mu_j)^2}}, \quad (5)$$

where $x_i(t)$ and $x_j(t)$ are the BOLD signals, and μ_i and μ_j are their time-averaged values. This approach is applied at the mesoscale with intermediate-frequency signals from 500 communities and at the macroscale with slow-frequency signals from 100 regions.

b) *Inter-module connectivity*: Inter-module connectivity quantifies the relationship between modules within each scale. It is based on the average BOLD signals from each module. The Pearson correlation between the average signals $\bar{x}_{m_i}$ and $\bar{x}_{m_j}$ for modules i and j is computed using the same formula as intra-module connectivity (see Equation (III-B4a)). These values populate the off-diagonal blocks of the global adjacency matrix.

c) *Differential thresholding*: Differential thresholding amplifies intra-module connectivity and attenuates inter-module interactions to highlight the brain's hierarchical structure. It applies distinct thresholds, θ_{intra} for intra-module and θ_{inter} for inter-module connections. These thresholds are determined through statistical analysis of connectivity patterns. The thresholded adjacency matrix $A_{thresh}^{(s)}$ is defined as:

$$A_{thresh}^{(s)} = \begin{cases} A_{i,j}^{(s)}, & \text{if } A_{i,j}^{(s)} \geq \theta_{intra} \text{ for } i = j \\ A_{i,j}^{(s)}, & \text{if } A_{i,j}^{(s)} \geq \theta_{inter} \text{ for } i \neq j \\ 0, & \text{otherwise} \end{cases} \quad (6)$$

This method retains significant intra- and inter-module connections while setting non-significant values to zero.

d) *Global adjacency matrix*: The global adjacency matrix consolidates module-specific connectivity matrices into a unified structure at each spatial scale. For each granularity level (microscale, mesoscale, macroscale), a distinct global adjacency matrix $A^{(s)}$ is created, where s denotes the scale. The matrix $A^{(s)}$ is $n \times n$, with n representing the number of nodes or communities at that scale. Diagonal blocks of $A^{(s)}$ represent intra-module connections, while off-diagonal blocks capture inter-module connections:

$$A^{(s)} = \begin{bmatrix} A_{1,1}^{(s)} & A_{1,2}^{(s)} & \cdots & A_{1,7}^{(s)} \\ A_{2,1}^{(s)} & A_{2,2}^{(s)} & \cdots & A_{2,7}^{(s)} \\ \vdots & \vdots & \ddots & \vdots \\ A_{7,1}^{(s)} & A_{7,2}^{(s)} & \cdots & A_{7,7}^{(s)} \end{bmatrix}, \quad (7)$$

where $A_{i,j}^{(s)}$ indicates the connectivity between modules i and j . This multiscale brain network dataset is crucial for precise training and analysis. All fMRI scans are processed to generate connectivity matrices at the microscale, mesoscale, and macroscale, which serve as the foundation for training and evaluating the BrainMGT framework.

C. Multiscale Graph Transformer

1) *Input Embeddings*: The BrainMGT model processes three adjacency matrices, representing multiscale brain networks at the micro, meso, and macro levels: A^{micro} , A^{meso} , and A^{macro} . Each matrix is $n \times n$, with n as the number of nodes or communities at the respective scale. Node features are derived from the rows of the adjacency matrix, where the feature vector for node i , $\mathbf{h}_i$, is $A[i, :]$. Edge features, representing connection strength, are given by matrix entries $A[i, j]$. Graph Attention Networks (GATs) [48] encode node and edge features using attention mechanisms to assess the

importance of neighboring nodes. The attention coefficient α_{ij} is computed as:

$$\alpha_{ij} = \frac{\exp(\text{LeakyReLU}(\mathbf{a}^\top [\mathbf{W}\mathbf{h}_i \oplus \mathbf{W}\mathbf{h}_j]))}{\sum_{k \in \mathcal{N}(i)} \exp(\text{LeakyReLU}(\mathbf{a}^\top [\mathbf{W}\mathbf{h}_i \oplus \mathbf{W}\mathbf{h}_k]))}, \quad (8)$$

where $\mathbf{W}$ is the weight matrix, $\mathbf{a}$ is the attention vector, and $\mathcal{N}(i)$ denotes the neighbors of node i . For each scale, separate GAT layers process the features, capturing both within- and between-module connectivity. The output embedding for node i , $\mathbf{h}'_i$, is computed by aggregating the features of neighboring nodes:

$$\mathbf{h}'_i = \sigma \left(\sum_{j \in \mathcal{N}(i)} \alpha_{ij} \mathbf{W}\mathbf{h}_j \right), \quad (9)$$

where σ is a non-linear activation function, such as ReLU.

2) *Positional Encoding*: Positional encoding is crucial for transformers to capture the relative arrangement of nodes in a graph. We use trainable positional embeddings, where each node i is assigned an embedding $\mathbf{p}_i$, optimized during training. These embeddings form a matrix P , with each row corresponding to a node's positional embedding. For a graph with n nodes, P is an $n \times d$ matrix, where d is the embedding dimension. The positional embedding $\mathbf{p}_i$ is added to the node's feature vector $\mathbf{h}_i$, producing the updated node feature vector $\mathbf{h}'_i$:

$$\mathbf{h}'_i = \mathbf{h}_i + \mathbf{p}_i \quad (10)$$

This method incorporates positional data into node features, improving the model's ability to interpret node relationships. In the BrainMGT framework, separate positional embeddings are created for each hierarchical scale (micro, meso, macro) to ensure accuracy at all levels.

3) *Scale-Specific Encoders*: The BrainMGT model uses distinct encoders for each hierarchical level to capture interactions across brain networks at different scales. Self-attention mechanisms are employed to assign importance to nodes and their edges, allowing the model to model complex relationships [49]. For a given node i , the attention weight α_{ij} is calculated based on its neighborhood $\mathcal{N}(i)$:

$$\alpha_{ij} = \frac{\exp(e_{ij})}{\sum_{k \in \mathcal{N}(i)} \exp(e_{ik})}, \quad (11)$$

where e_{ij} is the compatibility metric, $e_{ij} = \mathbf{q}_i^T \mathbf{k}_j$, and $\mathbf{q}_i = \mathbf{W}_q \mathbf{h}_i$, $\mathbf{k}_j = \mathbf{W}_k \mathbf{h}_j$. The output $\mathbf{h}'_i$ is a weighted sum of its neighbors' value vectors:

$$\mathbf{h}'_i = \sum_{j \in \mathcal{N}(i)} \alpha_{ij} \mathbf{v}_j, \quad (12)$$

with $\mathbf{v}_j = \mathbf{W}_v \mathbf{h}_j$. Multi-head attention is applied to capture diverse aspects of connectivity by running several attention heads concurrently. The final output for node i is obtained by concatenating the results from all heads and projecting them back to the original feature space:

$$\mathbf{h}''_i = \mathbf{W}_o [\mathbf{h}_i^{(1)} \parallel \mathbf{h}_i^{(2)} \parallel \cdots \parallel \mathbf{h}_i^{(H)}], \quad (13)$$

where H is the number of attention heads, and $\mathbf{W}_o$ is the output projection matrix. The values for H and d are set based on the scale and complexity of the network, typically with $H = 8$.

4) *Cross-Scale Encoders*: The cross-scale encoders in the BrainMGT model facilitate interactions between different brain scales, specifically between microscale (A^{micro}) and mesoscale (A^{meso}), and between mesoscale and macroscale (A^{macro}). These encoders process features from scale-specific encoders to model inter-scale dependencies. Cross-attention mechanisms compute attention scores based on the interaction between query vectors $\mathbf{Q}$ from one scale and key vectors $\mathbf{K}$ from another. For nodes i and j in different scales, the attention score α_{ij} is:

$$\alpha_{ij} = \frac{\exp(\mathbf{q}_i^\top \mathbf{k}_j)}{\sum_k \exp(\mathbf{q}_i^\top \mathbf{k}_k)}, \quad (14)$$

where $\mathbf{q}_i$ and $\mathbf{k}_j$ are the query and key vectors, respectively. These scores weigh the value vectors $\mathbf{V}$ from the target scale, producing the integrated representation:

$$\mathbf{h}'_i = \sum_j \alpha_{ij} \mathbf{v}_j. \quad (15)$$

The BrainMGT model uses two cross-scale encoders: one for microscale-mesoscale interactions ($C_{micro \leftrightarrow meso}$) and another for mesoscale-macroscale interactions ($C_{meso \leftrightarrow macro}$). This dual cross-attention approach captures hierarchical dependencies across scales, enhancing the model's multiscale network representation.

5) *Hierarchical Adaptive Fusion*: The hierarchical adaptive fusion integrates features from both scale-specific encoders (E_{micro} , E_{meso} , E_{macro}) and cross-scale encoders ($C_{micro \leftrightarrow meso}$, $C_{meso \leftrightarrow macro}$) to construct a unified brain network representation. The process involves aggregation and adaptive fusion. First, aggregation layers combine intra-scale features from the scale-specific encoders and inter-scale features from the cross-scale encoders. For node i , the aggregated feature vector $\mathbf{h}_i^{agg}$ is calculated by concatenating the scale-specific and cross-scale features:

$$\mathbf{h}_i^{agg} = [\mathbf{h}_i^{scale} || \mathbf{h}_i^{cross}]. \quad (16)$$

An adaptive fusion mechanism then assigns attention weights to the aggregated features, dynamically adjusting their contribution based on importance. The importance weight β_i for node i is computed using the softmax function:

$$\beta_i = \text{softmax}(\mathbf{W}_f \mathbf{h}_i^{agg}). \quad (17)$$

The final fused feature vector $\mathbf{h}_i^{fused}$ is obtained by element-wise multiplication:

$$\mathbf{h}_i^{fused} = \beta_i \odot \mathbf{h}_i^{agg}. \quad (18)$$

This fusion mechanism enables the model to prioritize relevant features for capturing hierarchical dependencies in the brain network, ensuring a comprehensive representation of multiscale dynamics.

6) *Classification*: In the final phase of the BrainMGT model, the fused node features $\mathbf{h}_i^{fused}$ are passed through multiple fully connected layers of a Multilayer Perceptron (MLP), followed by a softmax activation for classification

Algorithm 1 BrainMGT

Require: fMRI dataset $\mathbf{X} \in \mathbb{R}^{T \times V}$

Ensure: Disorder prediction $\hat{y}$

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1: procedure BRAINMGT( $\mathbf{X}$ )
2:   Construction of Multiscale Brain Networks:
3:    $\mathbf{X}_p \leftarrow \text{PreprocessfMRI}(\mathbf{X})$ 
4:   for each spatial scale  $s \in \{\text{Micro, Meso, Macro}\}$  do
5:      $\mathbf{S}_s \leftarrow \text{HierarchicalSpatialScales}(\mathbf{X}_p, s)$ 
6:      $\mathbf{T}_s \leftarrow \text{MultiBandTemporalScales}(\mathbf{S}_s)$ 
7:      $\mathcal{G}_s \leftarrow \text{MultiscaleConnectivity}(\mathbf{S}_s, \mathbf{T}_s)$ 
8:   end for
9:   Multiscale Graph Transformer:
10:  Initialize set of learnable parameters  $\theta$ 
11:  for epoch < epochs_total do
12:    for each spatial scale  $s \in \{\text{Micro, Meso, Macro}\}$  do
13:       $\mathbf{h}_i^s, \mathbf{e}_{ij}^s \leftarrow \text{InputEmbeddings}(\mathcal{G}_s)$ 
14:       $\mathbf{h}_G^s \leftarrow \text{ScaleSpecificEncoder}(\mathbf{h}_i^s, \mathbf{e}_{ij}^s)$ 
15:    end for
16:     $\mathbf{h}_{\text{cross}}^{\text{Micro, Meso}} \leftarrow \text{CrossScaleEncoder}(\mathbf{h}_G^{\text{Micro}}, \mathbf{h}_G^{\text{Meso}})$ 
17:     $\mathbf{h}_{\text{cross}}^{\text{Meso, Macro}} \leftarrow \text{CrossScaleEncoder}(\mathbf{h}_G^{\text{Meso}}, \mathbf{h}_G^{\text{Macro}})$ 
18:     $\mathbf{h}_{\text{fused}} \leftarrow \text{HierarchicalAdaptiveFusion}(\mathbf{h}_G^{\text{Micro}}, \mathbf{h}_{\text{cross}}^{\text{Micro, Meso}}, \mathbf{h}_G^{\text{Meso}}, \mathbf{h}_{\text{cross}}^{\text{Meso, Macro}}, \mathbf{h}_G^{\text{Macro}})$ 
19:     $\hat{y} \leftarrow \text{softmax}(\mathbf{W} \mathbf{h}_{\text{fused}} + \mathbf{b})$ 
20:    Calculate the loss function  $\mathcal{L}$ 
21:    Propagate the gradients backward and adjust the parameters  $\theta$ 
22:  end for
23:  return  $\hat{y}$ 
24: end procedure

```

into disease categories. For a node i in class c , the softmax output is:

$$P(y_i = c | \mathbf{h}_i^{fused}) = \frac{\exp(\mathbf{W}_c^T \mathbf{h}_i^{fused} + \mathbf{b}_c)}{\sum_k \exp(\mathbf{W}_k^T \mathbf{h}_i^{fused} + \mathbf{b}_k)}, \quad (19)$$

where $\mathbf{W}_c$ and $\mathbf{b}_c$ are the weight vector and bias for class c . This ensures the sum of the probabilities across all classes equals 1, forming a valid probability distribution. For binary classification, we use a cross-entropy loss function:

$$L_i = - \sum_c y_{i,c} \log P(y_i = c | \mathbf{h}_i^{fused}), \quad (20)$$

where $y_{i,c}$ is an indicator variable indicating the true class label. Minimizing this loss function improves classification accuracy. The Adam optimizer, with an adaptive learning rate, is used to minimize the loss, alongside a dynamic learning rate schedule to enhance convergence. The pseudocode for BrainMGT is provided in Algorithm 1.

IV. EXPERIMENTAL RESULTS

A. Experimental Settings

1) Datasets:

a) *Alzheimer's disease neuroimaging initiative (ADNI)*: The ADNI dataset⁵ supports AD progression studies, containing fMRI, PET, structural MRI, and genetic data. Participants

⁵<http://adni.loni.usc.edu/data-samples/access-data>

TABLE I
OVERVIEW OF DATASET STATISTICS

Dataset	Number of Subjects	Age Range (years)	Gender (M/F)	Class Labels (count)
ADNI	427	50 - 100	227 / 199	CN (280), AD (146)
PPMI	324	40 - 85	194 / 130	Control (94), PD (230)
ABIDE	1118	8 - 40	957 / 161	Control (581), ASD (537)

were selected based on fMRI availability and diagnostic labels of CN or AD, excluding those with significant comorbid conditions like stroke or major depression to reduce confounding factors. The dataset is used for binary classification to differentiate between cognitively normal (CN) subjects and those with AD. The cohort includes 426 individuals aged 50-100, consisting of 227 males and 199 females, with 280 labeled as CN and 146 as AD.

b) Parkinson's progression markers initiative (PPMI): The PPMI dataset⁶ includes fMRI data, DAT scans, and CSF biomarkers. It supports binary classification of Parkinson's disease (PD) versus control groups. Participants were selected based on fMRI scans and confirmed diagnoses of PD or healthy controls, excluding those with other neurological disorders or severe systemic illnesses to minimize biases. The cohort consists of 324 participants, aged 40-85, including 194 males and 130 females. The group is labeled as 94 controls and 230 PD subjects.

c) Autism brain imaging data exchange (ABIDE): The ABIDE dataset⁷ provides fMRI data focused on ASD, along with anatomical MRI and phenotypic information. The dataset supports binary classification to distinguish control subjects from those with ASD. Participants were selected based on available fMRI data and diagnostic labels, excluding individuals with severe comorbid psychiatric disorders or low-quality data to ensure integrity. A total of 1118 subjects were included, aged 8-40, with a male predominance (957 males, 161 females). The labels used for classification were Control (581) and ASD (537), supporting analysis in hierarchical multiscale brain network models. Dataset statistics are provided in Table I.

2) Baselines: We evaluate BrainMGT against a range of baseline methods, grouped into four categories: traditional machine learning, graph-based learning, and single- and multiscale brain network models. This comparison highlights the advantages and improvements of BrainMGT over existing approaches.

a) Conventional machine learning models: The initial set of baseline models includes Conventional machine learning algorithms such as Support Vector Machine (SVM) [50], Random Forest (RF) [51], and Multilayer Perceptron (MLP) [52], chosen for their proven utility in classification tasks. These models operate on single-scale brain networks derived from fMRI data, with the BrainGB pipeline used to preprocess the data, which involves steps such as motion correction, slice timing alignment, spatial normalization, smoothing, and artifact removal. Features based on graph theory are then extracted from these brain networks and fed into the models. The

models are implemented using the scikit-learn library, with grid search-based hyperparameter tuning applied to ensure optimized performance while maintaining default settings.

b) Graph learning models: The second category comprises advanced graph learning models, specifically Graph Convolutional Network (GCN) [53], Graph Isomorphism Network (GIN) [54], and Graph Attention Network (GAT) [48]. These architectures are selected due to their capacity to natively handle graph-structured data, making them particularly well-suited for brain network analysis tasks. As with traditional models, single-scale brain networks are generated from fMRI data using the BrainGB pipeline. However, unlike the conventional approach of feature extraction, these models utilize the entire brain network as input. The PyTorch Geometric framework is employed for model implementation and training, with cross-validation guiding hyperparameter tuning while maintaining default configuration settings.

c) Single-scale brain network models: The third category includes models designed specifically for single-scale brain network analysis, such as BrainGNN [28], BrainGB [5], and BRAINNETTF [29]. These architectures are chosen for their targeted design and demonstrated success in brain network analysis tasks. Brain networks are derived from fMRI data using the BrainGB pipeline and serve as input directly, bypassing the need for feature extraction or selection processes. Each model is trained in accordance with its own implementation protocols, ensuring consistency with its intended design and optimization strategies.

d) Multiscale brain network models: This category includes MMTGCN [11], MAHGCN [15], and Multi-atlas-GNN [16], leading models in handling multiple scales of brain network data. These models are relevant due to their hierarchical and multi-atlas strategies, similar to BrainMGT, which enhance sensitivity to pathological and normal variations across different brain scales. Implemented following their guidelines, all models use a multi-atlas-based approach to extract multiscale functional connectivity, enabling direct comparison of how effectively each integrates and interprets multiscale information relative to BrainMGT.

3) Implementation Details: The BrainMGT framework consists of two key components: the Hierarchical Multiscale Brain Networks Construction module and the Multiscale Brain Graph Transformer Model. It utilizes PyTorch Geometric for graph-based neural network development, and preprocessing of fMRI data is handled with the nibabel, Nilearn, and Nipype libraries. We use three datasets—ADNI, PPMI, and ABIDE—for training and testing, partitioning each into training, validation, and test subsets with an 80:10:10 ratio. Hyperparameter tuning was performed using grid search, optimizing parameters such as the learning rate (0.001), batch size (32), attention heads (8), dropout rate (0.5), and embedding dimension (128). The model is trained with the Adam optimizer for up to 200 epochs, with early stopping if validation loss plateaus for 10 epochs. The source code is publicly available on GitHub.⁸

⁶<https://www.ppmi-info.org/access-data-specimens/download-data>

⁷http://fcon_1000.projects.nitrc.org/indi/abide

⁸<https://github.com/Ahsan-Shehzad/BrainMGT>

TABLE II
PERFORMANCE COMPARISON (%). BRAINMGT EXHIBITS STATISTICALLY SIGNIFICANT ADVANCEMENTS OVER THE BASELINE MODELS, VALIDATED THROUGH T-TESTS (P-VALUE < 0.05)

Baseline Group	Method	ADNI			PPMI			ABIDE		
		Accuracy	F1	AUC	Accuracy	F1	AUC	Accuracy	F1	AUC
Machine Learning	SVM	55.64±4.83	50.04±2.74	63.06±1.88	72.02±1.90	38.65±1.02	40.19±1.56	61.95±2.44	41.44±2.78	65.33±0.84
	Random Forest	57.31±0.71	51.46±2.85	65.71±0.65	75.94±4.45	39.64±1.74	43.61±3.28	63.69±3.55	40.32±0.74	66.71±1.73
	MLP	56.55±3.82	50.74±4.08	65.37±2.25	76.01±1.15	37.81±2.78	42.93±0.61	65.44±3.73	42.54±4.74	67.12±2.93
Graph Learning	GCN	61.49±2.20	55.77±3.58	67.84±0.89	80.98±1.29	40.40±2.53	49.45±4.92	69.39±4.69	49.68±2.23	71.51±3.80
	GIN	63.01±4.58	57.74±4.98	69.06±3.88	80.42±2.64	42.21±4.42	50.61±4.38	70.35±4.38	50.17±3.02	73.12±1.68
	GAT	62.58±3.79	56.52±4.85	68.89±4.09	81.50±1.19	42.76±1.72	52.86±0.63	71.76±2.88	52.61±3.19	75.77±2.35
Single-scale Brain Network Models	BrainGNN	66.34±2.50	61.11±4.12	70.06±3.46	82.23±4.46	45.26±2.73	55.77±1.87	75.96±1.47	53.19±4.35	76.74±3.59
	BrainGB	67.03±3.93	62.06±3.13	71.63±2.03	81.16±3.33	46.35±2.27	55.50±4.84	76.16±3.44	54.79±4.32	78.30±0.51
	BRAINNETTF	68.03±0.88	63.36±2.28	74.91±4.96	82.83±1.88	48.38±4.63	57.68±2.05	77.26±4.63	57.02±3.87	79.70±4.58
Multiscale Brain Network Models	MMTGCN	71.23±1.79	68.57±1.14	74.91±4.96	84.42±3.05	55.53±4.29	63.93±4.97	83.73±2.62	63.88±3.52	86.41±0.64
	MAHGCN	70.92±1.65	66.67±0.54	73.17±4.77	85.23±1.18	59.47±1.23	64.13±4.49	82.44±1.78	64.66±1.52	85.61±4.00
	Multi-atlas-GNN	73.78±4.76	70.20±3.84	78.90±2.77	87.99±2.24	58.37±1.28	65.07±3.55	84.11±2.52	66.39±4.69	86.57±0.58
Ours	BrainMGT	76.71±1.47	73.51±3.91	82.02±4.32	86.29±1.19	62.90±4.78	69.12±1.29	86.14±4.91	68.56±1.80	89.82±1.66

4) *Evaluation Metrics*: We evaluate the BrainMGT framework's performance using accuracy, F1 score, Area Under the Receiver Operating Characteristic Curve (AUC), and the paired t-test. Accuracy reflects overall prediction correctness, while the F1 score balances precision and recall, crucial for imbalanced data. This is particularly important in brain disease diagnosis, where false positives and negatives have significant consequences. AUC measures the model's ability to distinguish between classes, essential for identifying subtle brain condition differences. The paired t-test compares performance improvements over baseline methods. These metrics are also applied to baseline methods for a comprehensive comparison.

B. Comparison With Baseline Methods

Table II shows that BrainMGT performs consistently across all three datasets, with variations in accuracy, F1-score, and AUC. For the ADNI dataset, BrainMGT achieves an accuracy of 76.71±1.47, F1-score of 73.51±3.91, and AUC of 82.02±4.32, effectively distinguishing Alzheimer's disease (AD) from cognitively normal (CN) subjects. In the PPMI dataset, focusing on Parkinson's disease (PD) and healthy controls, it attains an accuracy of 86.29±1.19, F1-score of 62.90±4.78, and AUC of 69.12±1.29, indicating a sensitivity-precision trade-off. For the ABIDE dataset, focused on Autism Spectrum Disorder (ASD), BrainMGT achieves an accuracy of 86.14±4.91, F1-score of 68.56±1.80, and AUC of 89.89±1.66, demonstrating its robustness across diverse brain conditions.

1) *Conventional Machine Learning Models*: Traditional machine learning models, such as SVM, RF, and MLP, use graph-theory-based features from single-scale brain networks. These models struggle to capture the hierarchical and multiscale nature of brain connectivity. On the ADNI dataset, Random Forest achieves 57.31% accuracy, 51.46% F1 score, and 65.71% AUC, while BrainMGT outperforms it by 33.85% in accuracy, 42.05% in F1 score, and 24.76% in AUC. On the PPMI dataset, Random Forest achieves 75.94% accuracy, 39.64% F1 score, and 43.61% AUC, with BrainMGT improving accuracy by 10.36%, F1 by 58.65%, and AUC by

25.51%. On the ABIDE dataset, MLP reaches 65.44% accuracy, 42.54% F1 score, and 67.12% AUC, while BrainMGT improves accuracy by 31.54%, F1 by 61.02%, and AUC by 33.88%. BrainMGT's advantage arises from its ability to model hierarchical interactions across multiple scales, enhanced by Graph Attention Networks (GATs) and positional encoding, crucial for accurate diagnosis.

2) *Graph Learning Models*: Graph-based models, such as GCN, GIN, and GAT, operate on brain network graphs derived from single-scale data. However, they struggle to integrate information across multiple scales. On the ADNI dataset, the GIN model achieves 63.01% accuracy, 57.74% F1 score, and 69.06% AUC, while BrainMGT outperforms it by 21.78% in accuracy, 27.31% in F1 score, and 18.77% in AUC. On the PPMI dataset, GAT achieves 81.50% accuracy, 42.76% F1 score, and 52.86% AUC, with BrainMGT improving these by 5.88%, 47.03%, and 30.75%, respectively. For the ABIDE dataset, GAT leads with 71.76% accuracy, 52.61% F1 score, and 75.77% AUC, but BrainMGT surpasses it by 20.05% in accuracy, 30.31% in F1 score, and 18.61% in AUC. BrainMGT's advantage lies in its multiscale approach, integrating information across brain levels through scale-specific encoders and hierarchical adaptive fusion.

3) *Single-Scale Brain Network Models*: Models such as BrainGNN, BrainGB, and BRAINNETTF focus on single-scale brain networks, which limits their ability to capture complex brain functions. On the ADNI dataset, BRAINNETTF achieves 68.03% accuracy, 63.36% F1 score, and 74.91% AUC. BrainMGT outperforms it by 12.74% in accuracy, 15.99% in F1 score, and 9.52% in AUC. On the PPMI dataset, BRAINNETTF reaches 82.83% accuracy, 48.38% F1 score, and 57.68% AUC, while BrainMGT improves by 4.18%, 30.00%, and 19.89%, respectively. For the ABIDE dataset, BRAINNETTF performs at 77.26% accuracy, 57.02% F1 score, and 79.70% AUC, but BrainMGT surpasses it by 11.47% in accuracy, 20.21% in F1 score, and 12.77% in AUC. The limited scope of single-scale models contributes to their lower performance compared to BrainMGT, which leverages multiscale networks and hierarchical encoders for

TABLE III
PERFORMANCE IMPACT OF REMOVING INDIVIDUAL COMPONENTS IN BRAINMGT

Component Ablated	ADNI			PPMI			ABIDE		
	Accuracy	F1-Score	AUC	Accuracy	F1-Score	AUC	Accuracy	F1-Score	AUC
None (Full BrainMGT)	76.71±1.47	73.51±3.91	82.02±4.32	86.29±1.19	62.90±4.78	69.12±1.29	86.14±4.91	68.56±1.80	89.82±1.66
Preprocessing	61.08±3.71	58.37±3.91	64.97±3.68	68.42±4.49	49.69±1.74	55.40±3.11	68.18±1.89	54.33±1.64	71.11±3.63
Hierarchical Spatial Scales	67.17±1.18	64.06±1.67	71.85±0.75	75.44±2.79	54.94±4.27	60.16±2.14	75.04±2.79	59.69±1.77	78.28±4.30
Multi-band Temporal Scales	73.82±4.23	70.71±1.56	78.83±0.73	82.93±1.13	60.73±2.68	66.43±2.54	82.87±3.69	66.03±1.62	86.49±1.75
Modular Connectivity	70.64±2.55	67.78±3.26	75.61±2.54	79.76±4.51	57.95±1.56	63.82±3.20	79.63±1.51	63.11±3.95	83.05±0.95
Scale-specific Encoders	68.04±3.57	64.89±4.41	72.35±4.60	76.13±1.76	55.40±3.11	55.40±3.11	75.92±4.71	60.45±0.62	79.31±3.25
Cross-scale Encoders	72.14±2.60	69.47±2.60	77.44±4.67	81.30±1.01	59.17±3.19	64.98±2.86	81.14±0.73	64.45±3.77	84.56±4.72
Hierarchical Adaptive Fusion	70.04±1.43	66.93±3.41	74.72±4.42	78.59±1.39	57.46±0.76	63.06±1.08	78.63±1.54	62.50±2.19	82.26±2.65

improved predictive accuracy and a deeper understanding of brain disorders.

4) *Multiscale Brain Network Models*: This group includes advanced models such as MMTGCN, MAHGCN, and Multi-atlas-GNN, which operate on multiscale brain networks and are key competitors to BrainMGT. On the ADNI dataset, Multi-atlas-GNN achieves 73.78% accuracy, 70.20% F1 score, and 78.90% AUC, while BrainMGT exceeds it by 3.97%, 4.72%, and 4.12%, respectively. On the PPMI dataset, Multi-atlas-GNN reports 87.99% accuracy, 58.37% F1 score, and 65.07% AUC, with BrainMGT outperforming it in F1 score by 7.77% and AUC by 6.21%. In the ABIDE dataset, Multi-atlas-GNN achieves 84.11% accuracy, 66.39% F1 score, and 86.57% AUC, but BrainMGT improves by 2.41%, 3.27%, and 3.84%, respectively. While Multi-atlas-GNN performs well, BrainMGT's hierarchical fusion and cross-scale encoders provide superior feature integration, leading to better predictive performance and a deeper understanding of multiscale brain networks.

C. Ablation Study

We conduct ablation studies to assess the contribution of each component in the BrainMGT framework. For each experiment, we sequentially remove a specific component and evaluate its impact on model performance. First, we bypass preprocessing and use raw fMRI data to examine the effect of noise reduction and data quality. Second, we exclude multiscale features, cross-scale encoders, and hierarchical fusion mechanisms by limiting the model to a single spatial resolution. Third, we restrict temporal frequencies to the traditional range (0.01–0.1 Hz) to evaluate the role of frequency processing. Fourth, we remove the brain network's modular organization to test the importance of modular connectivity. Fifth, we exclude scale-specific encoders to assess intra-scale feature extraction. Sixth, we omit cross-scale encoders to examine inter-scale integration. Finally, we exclude hierarchical fusion to determine its role in multiscale integration. These experiments are conducted on ADNI, PPMI, and ABIDE datasets with consistent data splits. Performance is evaluated using accuracy, F1-score, and AUC, with statistical significance validated through multiple iterations. Table III summarizes the impact of component removal on performance.

The performance of each ablated model is compared to the full BrainMGT model. Omitting preprocessing causes the largest accuracy drop: 15.63% on ADNI, 17.87% on PPMI, and 17.96% on ABIDE, highlighting the importance

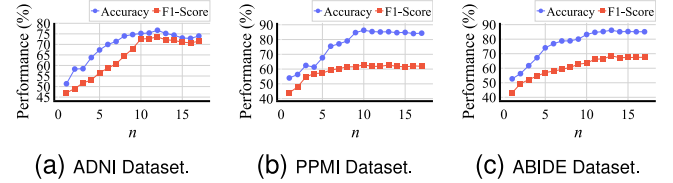


Fig. 2. The accuracy and F1-score of BrainMGT w.r.t. different n values.

of noise reduction and data quality. Removing hierarchical spatial scales leads to accuracy declines of 9.54%, 10.85%, and 11.10% on ADNI, PPMI, and ABIDE, respectively, emphasizing the role of multiscale spatial information. Excluding multi-band temporal scales results in a 3% accuracy drop, indicating that temporal frequency processing is important but less critical than spatial scales. Omitting modular connectivity reduces accuracy by 6.5%, stressing the importance of brain network modularity. Removing scale-specific encoders causes an 8-10% drop in accuracy, highlighting the significance of intra-scale feature extraction. Omitting cross-scale encoders results in a 4-5% accuracy decline, showing that inter-scale integration is important but slightly less so. Finally, excluding hierarchical adaptive fusion reduces accuracy by 7%, underscoring its key role in integrating multiscale information.

D. Parameter Analysis

This section analyzes key parameters in the BrainMGT framework: the number of functional brain modules (n) and the number of layers (l). These parameters directly affect diagnostic accuracy. We evaluate combinations of n and l to identify configurations that optimize performance across datasets. The parameter n represents the number of functional networks from the Schaefer atlas, varied from 1 to 17 to assess the impact of modular granularity on network construction. The parameter l defines the model depth, tested between 1 and 5 layers, to evaluate the trade-off between shallow and deep architectures. Both parameters are crucial for modeling hierarchical brain connectivity and optimizing performance. We selected initial values for n and l based on preliminary experiments and performed a grid search to evaluate all combinations. Performance was measured using accuracy and F1-Score on the ADNI, PPMI, and ABIDE datasets.

The results, summarized in Figs. 2 and 3, highlight clear trends. For the ADNI dataset, accuracy peaks at $n = 12$ modules with 76.71% accuracy and a 73.51% F1-Score. The optimal depth is $l = 2$, with diminishing returns beyond this

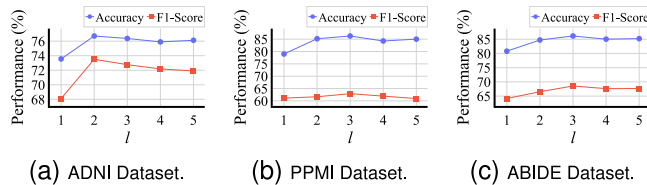


Fig. 3. The accuracy and F1-score of BrainMGT w.r.t. different l values.

point. In the PPMI dataset, performance is highest at $n = 10$ modules, achieving 86.29% accuracy and a 62.90% F1-Score. The best depth is $l = 3$. For the ABIDE dataset, $n = 13$ and $l = 3$ yield 86.14% accuracy and a 68.56% F1-Score. Increasing n enhances performance by providing finer connectivity details but introduces overfitting beyond optimal values. Similarly, moderate depth (2-3 layers) effectively captures hierarchical interactions, while additional layers yield limited benefits. These findings suggest that optimal parameter values balance granularity and complexity to maximize diagnostic performance.

E. Discussion

The BrainMGT framework demonstrates strong capabilities in diagnosing brain disorders through multiscale brain network analysis. Its consistent performance across datasets like ADNI, PPMI, and ABIDE shows its ability to capture complex hierarchical brain interactions essential for accurate diagnosis. BrainMGT outperforms conventional machine learning and state-of-the-art graph models, especially in F1-score and AUC, highlighting its superior balance between sensitivity and specificity. A critical evaluation reveals that BrainMGT's multiscale approach, including hierarchical adaptive fusion and cross-scale encoders, enhances its sensitivity to brain connectivity nuances across scales. This approach supports the knowledge that brain structure is defined in fractal terms. It offers a more comprehensive representation of brain function, which is vital for diagnosing conditions such as Alzheimer's and Parkinson's disease. This approach has the potential for being considered for other biological systems that have fractal properties.

1) *Limitations and Future Work:* BrainMGT presents several technical and methodological limitations. A primary constraint is its reliance on large-scale, high-quality fMRI datasets, which may not be accessible in all clinical settings, reducing generalizability. Although BrainMGT performs well on selected datasets, its applicability to other brain conditions beyond Alzheimer's, Parkinson's, and autism spectrum disorders remains untested. Furthermore, the model is evaluated on only three datasets (ADNI, PPMI, and ABIDE). We will test it on additional datasets, such as REST-METAMDD, OASIS, ADHD-200, and WMH, to assess its robustness across diverse neurological conditions. We require more diverse datasets with varying demographic and clinical characteristics to validate the model's broader applicability. The computational complexity of BrainMGT, including the high training time and resource requirements, can hinder real-world implementation. Additionally, the absence of qualitative analysis and model visualizations limits interpretability. We will

explore visualizations to improve understanding of the model decision-making process, providing valuable clinical insights. Integrating BrainMGT with smart health systems, including wearable devices and home health monitoring tools, offers a promising avenue for real-time brain health monitoring. Future developments will adapt BrainMGT for scalability and compatibility with these technologies, facilitating its deployment in daily health management and expanding its accessibility. Additionally, optimizing BrainMGT for real-time processing is a priority to enhance its usability in dynamic consumer applications such as neurofeedback or brain-computer interfaces (BCI). Future work will focus on reducing computational latency and improving processing efficiency to facilitate real-time deployment. Addressing these limitations is essential to improve the practicality of the model and promote its adoption in clinical neuroimaging.

V. CONCLUSION

In this study, we have addressed the limitations of existing multi-atlas methods for modeling brain networks, which often fail to represent the hierarchical organization of brain connectivity at multiple spatial and temporal scales. To overcome this issue, we developed the Multiscale Brain Graph Transformer (BrainMGT) model, which constructs multiscale brain networks to better capture the hierarchical structure of brain function. BrainMGT models connectivity across three spatial scales, microscale, mesoscale, and macroscale, while preserving the brain's modular organization. It also incorporates multiple temporal scales, rather than relying on a single temporal scale from BOLD signals. Leveraging self-attention and cross-attention mechanisms, the BrainMGT framework effectively extracts and integrates features from both intra-scale and inter-scale data, thereby enhancing diagnostic precision for brain disorders. We validated the model on three real-world fMRI datasets, i.e., ADNI, PPMI, and ABIDE, where it outperformed existing techniques in the diagnosis of neurological conditions. These results underscore the potential of BrainMGT to offer a more detailed and accurate representation of brain connectivity with the potential for many biomedical applications that also have such hierarchical structures.

REFERENCES

- [1] M. Hallett et al., "Human brain connectivity: Clinical applications for clinical neurophysiology," *Clin. Neurophysiol.*, vol. 131, no. 7, pp. 1621–1651, Jul. 2020, doi: [10.1016/j.clinph.2020.03.031](https://doi.org/10.1016/j.clinph.2020.03.031).
- [2] Q. Yu et al., "Application of graph theory to assess static and dynamic brain connectivity: Approaches for building brain graphs," *Proc. IEEE*, vol. 106, no. 5, pp. 886–906, May 2018, doi: [10.1109/JPROC.2018.2825200](https://doi.org/10.1109/JPROC.2018.2825200).
- [3] A. Ashraf, Z. Qingjie, W. H. K. Bangyal, and M. Iqbal, "Analysis of brain imaging data for the detection of early age autism spectrum disorder using transfer learning approaches for Internet of Things," *IEEE Trans. Consum. Electron.*, vol. 70, no. 1, pp. 4478–4489, Feb. 2024, doi: [10.1109/tce.2023.3328479](https://doi.org/10.1109/tce.2023.3328479).
- [4] M. R. Ahmed, Y. Zhang, Z. Feng, B. Lo, O. T. Inan, and H. Liao, "Neuroimaging and machine learning for dementia diagnosis: Recent advancements and future prospects," *IEEE Rev. Biomed. Eng.*, vol. 12, pp. 19–33, 2019, doi: [10.1109/RBME.2018.2886237](https://doi.org/10.1109/RBME.2018.2886237).
- [5] H. Cui et al., "BrainGB: A benchmark for brain network analysis with graph neural networks," *IEEE Trans. Med. Imag.*, vol. 42, no. 2, pp. 493–506, Feb. 2023, doi: [10.1109/TMI.2022.3218745](https://doi.org/10.1109/TMI.2022.3218745).

- [6] J. Yang, L. Li, L. Y. Por, S. Bourouis, S. Dhahbi, and A. Ayub Khan, "Harnessing multimodal data and deep learning for comprehensive gait analysis in pediatric cerebral palsy," *IEEE Trans. Consum. Electron.*, vol. 70, no. 3, pp. 5401–5410, Aug. 2024, doi: [10.1109/tce.2024.3482689](https://doi.org/10.1109/tce.2024.3482689).
- [7] Z. Wang, J. Xin, Z. Wang, Y. Yao, Y. Zhao, and W. Qian, "Brain functional network modeling and analysis based on fMRI: A systematic review," *Cogn. Neurodyn.*, vol. 15, no. 3, pp. 389–403, Jun. 2021, doi: [10.1007/s11571-020-09630-5](https://doi.org/10.1007/s11571-020-09630-5).
- [8] E. Bullmore and O. Sporns, "Complex brain networks: Graph theoretical analysis of structural and functional systems," *Nat. Rev. Neurosci.*, vol. 10, no. 3, pp. 186–198, Feb. 2009, doi: [10.1038/nrn2575](https://doi.org/10.1038/nrn2575).
- [9] A. Bessadok, M. A. Mahjoub, and I. Rekik, "Graph neural networks in network neuroscience," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 45, no. 5, pp. 5833–5848, May 2023, doi: [10.1109/TPAMI.2022.3209686](https://doi.org/10.1109/TPAMI.2022.3209686).
- [10] H. Tang et al., "A comprehensive survey of complex brain network representation," *Meta-Radiol.*, vol. 1, no. 3, Nov. 2023, Art. no. 100046, doi: [10.1016/j.metrad.2023.100046](https://doi.org/10.1016/j.metrad.2023.100046).
- [11] D. Yao et al., "A mutual multi-scale triplet graph convolutional network for classification of brain disorders using functional or structural connectivity," *IEEE Trans. Med. Imag.*, vol. 40, no. 4, pp. 1279–1289, Apr. 2021, doi: [10.1109/tmi.2021.3051604](https://doi.org/10.1109/tmi.2021.3051604).
- [12] A. Shehzad et al., "Graph transformers: A survey," 2024, *arXiv:2407.09777*.
- [13] C. Ying et al., "Do transformers really perform badly for graph representation?" in *Proc. Adv. Neural Inf. Process. Syst.*, vol. 34, 2021, pp. 28877–28888.
- [14] D. Kreuzer, D. Beaini, W. Hamilton, V. Létourneau, and P. Tossou, "Rethinking graph transformers with spectral attention," in *Proc. 35th Int. Conf. Neural Inf. Process. Syst.*, vol. 34, 2021, pp. 21618–21629.
- [15] M. Liu, H. Zhang, F. Shi, and D. Shen, "Hierarchical graph convolutional network built by multiscale atlases for brain disorder diagnosis using functional connectivity," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 35, no. 11, pp. 15182–15194, Nov. 2024, doi: [10.1109/tnnls.2023.3282961](https://doi.org/10.1109/tnnls.2023.3282961).
- [16] D.-J. Lee et al., "Spectral graph neural network-based multi-atlas brain network fusion for major depressive disorder diagnosis," *IEEE J. Biomed. Health Inform.*, vol. 28, no. 5, pp. 2967–2978, May 2024, doi: [10.1109/jbhi.2024.3366662](https://doi.org/10.1109/jbhi.2024.3366662).
- [17] R. F. Betzel and D. S. Bassett, "Multi-scale brain networks," *NeuroImage*, vol. 160, pp. 73–83, Oct. 2017, doi: [10.1016/j.neuroimage.2016.11.006](https://doi.org/10.1016/j.neuroimage.2016.11.006).
- [18] Y. Du, S. Fang, X. He, and V. D. Calhoun, "A survey of brain functional network extraction methods using fMRI data," *Trends Neurosci.*, vol. 47, no. 8, pp. 608–621, 2024, doi: [10.1016/j.tins.2024.05.011](https://doi.org/10.1016/j.tins.2024.05.011).
- [19] B. Hu, Z.-H. Guan, G. Chen, and C. L. P. Chen, "Neuroscience and network dynamics toward brain-inspired intelligence," *IEEE Trans. Cybern.*, vol. 52, no. 10, pp. 10214–10227, Oct. 2022, doi: [10.1109/tycb.2021.3071110](https://doi.org/10.1109/tycb.2021.3071110).
- [20] E. D'Angelo and V. Jirsa, "The quest for multiscale brain modeling," *Trends Neurosci.*, vol. 45, no. 10, pp. 777–790, Oct. 2022, doi: [10.1016/j.tins.2022.06.007](https://doi.org/10.1016/j.tins.2022.06.007).
- [21] C. Peng et al., "mBGT: Encoding brain signals with multimodal brain graph transformer," *IEEE Trans. Consum. Electron.*, early access, Jul. 9, 2024, doi: [10.1109/tce.2024.3425478](https://doi.org/10.1109/tce.2024.3425478).
- [22] A. Shehzad, D. Zhang, S. Yu, S. Abid, and F. Xia, "Dynamic graph transformer for brain disorder diagnosis," *IEEE J. Biomed. Health Inform.*, early access, Feb. 5, 2025, doi: [10.1109/jbhi.2025.3538040](https://doi.org/10.1109/jbhi.2025.3538040).
- [23] D. Sadeghi et al., "An overview of artificial intelligence techniques for diagnosis of schizophrenia based on magnetic resonance imaging modalities: Methods, challenges, and future works," *Comput. Biol. Med.*, vol. 146, Jul. 2022, Art. no. 105554, doi: [10.1016/j.combiomed.2022.105554](https://doi.org/10.1016/j.combiomed.2022.105554).
- [24] J. Kawahara et al., "BrainNetCNN: Convolutional neural networks for brain networks; towards predicting neurodevelopment," *NeuroImage*, vol. 146, pp. 1038–1049, Feb. 2017, doi: [10.1016/j.neuroimage.2016.09.046](https://doi.org/10.1016/j.neuroimage.2016.09.046).
- [25] W. Tong et al., "fMRI-based brain disease diagnosis: A graph network approach," *IEEE Trans. Med. Robot. Bionics*, vol. 5, no. 2, pp. 312–322, May 2023, doi: [10.1109/TMRB.2023.3270481](https://doi.org/10.1109/TMRB.2023.3270481).
- [26] J. Lyu, X. Zhao, and M. S. Hossain, "Temporal super-resolution in T1 mapping: Pioneering speed and detail preservation for multimodal consumer health data," *IEEE Trans. Consum. Electron.*, vol. 70, no. 4, pp. 7195–7202, Nov. 2024, doi: [10.1109/tce.2024.3379482](https://doi.org/10.1109/tce.2024.3379482).
- [27] A. D. Arya et al., "A systematic review on machine learning and deep learning techniques in the effective diagnosis of Alzheimer's disease," *Brain Informat.*, vol. 10, no. 1, p. 17, Dec. 2023, doi: [10.1186/s40708-023-00195-7](https://doi.org/10.1186/s40708-023-00195-7).
- [28] X. Li et al., "BrainGNN: Interpretable brain graph neural network for fMRI analysis," *Med. Image Anal.*, vol. 74, Dec. 2021, Art. no. 102233, doi: [10.1016/j.media.2021.102233](https://doi.org/10.1016/j.media.2021.102233).
- [29] X. Kan, W. Dai, H. Cui, Z. Zhang, Y. Guo, and C. Yang, "Brain network transformer," in *Proc. Adv. Neural Inf. Process. Syst. (NeurIPS)*, Nov. 2022, pp. 1–23.
- [30] B. Min et al., "Recent advances in natural language processing via large pre-trained language models: A survey," *ACM Comput. Surv.*, vol. 56, no. 2, pp. 1–40, Sep. 2023, doi: [10.1145/3605943](https://doi.org/10.1145/3605943).
- [31] S. Khan, M. Naseer, M. Hayat, S. W. Zamir, F. S. Khan, and M. Shah, "Transformers in vision: A survey," *ACM Comput. Surv.*, vol. 54, no. 10s, pp. 1–41, 2022, doi: [10.1145/3505244](https://doi.org/10.1145/3505244).
- [32] D. Zhang, M. Zhang, H. Zhang, L. Yang, and H. Lin, "MultiMET: A multimodal dataset for metaphor understanding," in *Proc. 59th Annu. Meeting Assoc. Comput. Linguist. 11th Int. Joint Conf. Nat. Lang. Process.*, Aug. 2021, pp. 3214–3225, doi: [10.18653/v1/2021.acl-long.249](https://doi.org/10.18653/v1/2021.acl-long.249).
- [33] K. Yang, H. Zhang, J. Shi, and J. Ma, "BANDT: A border-aware network with deformable transformers for visual tracking," *IEEE Trans. Consum. Electron.*, vol. 69, no. 3, pp. 377–390, Aug. 2023, doi: [10.1109/tce.2023.3251407](https://doi.org/10.1109/tce.2023.3251407).
- [34] S. Yu, S. Jin, M. Li, T. Sarwar, and F. Xia, "Long-range brain graph transformer," in *Proc. 38th Annu. Conf. Neural Inf. Process. Syst.*, 2024, pp. 1–17.
- [35] D. K. Kumar, S. P. Arjunan, and B. Aliahmad, *Fractals: Applications in Biological Signalling and Image Processing*. Boca Raton, FL, USA: CRC Press, Feb. 2017, doi: [10.1201/9781315165868](https://doi.org/10.1201/9781315165868).
- [36] D. S. Bassett and O. Sporns, "Network neuroscience," *Nat. Neurosci.*, vol. 20, no. 3, pp. 353–364, Feb. 2017, doi: [10.1038/nrn.4502](https://doi.org/10.1038/nrn.4502).
- [37] H. S. Seung, "Reading the book of memory: Sparse sampling versus dense mapping of connectomes," *Neuron*, vol. 62, no. 1, pp. 17–29, Apr. 2009, doi: [10.1016/j.neuron.2009.03.020](https://doi.org/10.1016/j.neuron.2009.03.020).
- [38] O. Sporns, "Structure and function of complex brain networks," *Dialogues Clin. Neurosci.*, vol. 15, no. 3, pp. 247–262, Sep. 2013, doi: [10.31887/dcms.2013.15.3/osporns](https://doi.org/10.31887/dcms.2013.15.3/osporns).
- [39] Y. He and A. Evans, "Graph theoretical modeling of brain connectivity," *Curr. Opin. Neurol.*, vol. 23, no. 4, pp. 341–350, Aug. 2010, doi: [10.1097/wco.0b013e32833aa567](https://doi.org/10.1097/wco.0b013e32833aa567).
- [40] L. Q. Uddin, K. Supekar, and V. Menon, "Reconceptualizing functional brain connectivity in autism from a developmental perspective," *Front. Hum. Neurosci.*, vol. 7, p. 458, Aug. 2013, doi: [10.3389/fnhum.2013.00458](https://doi.org/10.3389/fnhum.2013.00458).
- [41] A. Schaefer et al., "Local-global parcellation of the human cerebral cortex from intrinsic functional connectivity MRI," *Cerebral Cortex*, vol. 28, no. 9, pp. 3095–3114, Sep. 2018, doi: [10.1093/cercor/bhx179](https://doi.org/10.1093/cercor/bhx179).
- [42] V. D. Blondel, J.-L. Guillaume, R. Lambiotte, and E. Lefebvre, "Fast unfolding of communities in large networks," *J. Stat. Mech., Theory Exp.*, vol. 2008, no. 10, Oct. 2008, Art. no. P10008, doi: [10.1088/1742-5468/2008/10/p10008](https://doi.org/10.1088/1742-5468/2008/10/p10008).
- [43] X.-N. Zuo et al., "The oscillating brain: Complex and reliable," *NeuroImage*, vol. 49, no. 2, pp. 1432–1445, Jan. 2010, doi: [10.1016/j.neuroimage.2009.09.037](https://doi.org/10.1016/j.neuroimage.2009.09.037).
- [44] A. T. Baria, M. N. Baliki, T. Parrish, and A. V. Apkarian, "Anatomical and functional assemblies of brain bold oscillations," *J. Neurosci.*, vol. 31, no. 21, pp. 7910–7919, May 2011, doi: [10.1523/jneurosci.1296-11.2011](https://doi.org/10.1523/jneurosci.1296-11.2011).
- [45] J. E. Chen and G. H. Glover, "Bold fractional contribution to resting-state functional connectivity above 0.1 Hz," *NeuroImage*, vol. 107, pp. 207–218, Feb. 2015, doi: [10.1016/j.neuroimage.2014.12.012](https://doi.org/10.1016/j.neuroimage.2014.12.012).
- [46] S. Mallat, "A theory for multiresolution signal decomposition: The wavelet representation," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 11, no. 7, pp. 674–693, Jul. 1989, doi: [10.1109/34.192463](https://doi.org/10.1109/34.192463).
- [47] D. Cordes et al., "Frequencies contributing to functional connectivity in the cerebral cortex in 'resting-state' data," *Amer. J. Neuroradiol.*, vol. 22, no. 7, pp. 1326–1333, 2001.
- [48] P. Velickovic, G. Cucurull, A. Casanova, A. Romero, P. Liò, and Y. Bengio, "Graph attention networks," in *Proc. 6th Int. Conf. Learn. Represent. (ICLR)*, Apr. 2018, pp. 1–12. [Online]. Available: <https://doi.org/10.48550/arxiv.1710.10903>
- [49] A. Vaswani et al., "Attention is all you need," in *Proc. 31st Int. Conf. Neural Inf. Process. Syst.*, 2017, pp. 6000–6010.
- [50] C. Cortes and V. Vapnik, "Support-vector networks," *Mach. Learn.*, vol. 20, no. 3, pp. 273–297, Sep. 1995, doi: [10.1007/bf00994018](https://doi.org/10.1007/bf00994018).
- [51] L. Breiman, "Random forests," *Mach. Learn.*, vol. 45, no. 1, pp. 5–32, 2001, doi: [10.1023/A:1010933404324](https://doi.org/10.1023/A:1010933404324).

- [52] D. E. Rumelhart, G. E. Hinton, and R. J. Williams, "Learning representations by back-propagating errors," *Nature*, vol. 323, no. 6088, pp. 533–536, Oct. 1986, doi: [10.1038/323533a0](https://doi.org/10.1038/323533a0).
- [53] T. N. Kipf and M. Welling, "Semi-supervised classification with graph convolutional networks," in *Proc. 5th Int. Conf. Learn. Represent.*, Apr. 2017, pp. 1–14.
- [54] K. Xu, W. Hu, J. Leskovec, and S. Jegelka, "How powerful are graph neural networks?" in *Proc. Int. Conf. Learn. Represent. (ICLR)*, New Orleans, LA, USA, May 2019, pp. 1–17.



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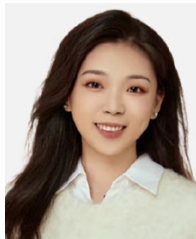
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